CYBERSECURITY & ETHICAL HACKING INTERNSHIP PROGRAM

Foundation & Environment Setup TASK 1

Lab Setup Report & Study Notes

Days 1 – 12 ApexPlanet Software Pvt. Ltd.

Program	Cybersecurity & Ethical Hacking Internship
Task	Task 1 – Foundation & Environment Setup
Timeline	Days 1–12
Organization	ApexPlanet Software Pvt. Ltd.
Tools Used	VirtualBox, Kali Linux, Metasploitable2, Wireshark, Nmap, Burp Suite, OpenSSL

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1.1 The CIA Triad

The CIA Triad is the foundational model for information security, representing three core principles that guide security policies and controls in any organization.

Principle	Definition	Example Controls
Confidentiality	Ensures that information is only accessible to those authorized to access it.	Encryption, Access Control Lists, MFA
Integrity	Guarantees that data is accurate, complete, and has not been tampered with.	Hashing (SHA256), Digital Signatures, Checksums
Availability	Ensures that systems and data are accessible to authorized users when needed.	Backups, Redundancy, DDoS Protection, Failover

1.2 Common Threat Types

Threat	Description
Phishing	Deceptive emails or websites that trick users into revealing credentials or installing malware.
Malware	Malicious software including viruses, trojans, worms, and spyware designed to damage or steal data.
DDoS (Distributed Denial of Service)	Flooding a server or network with traffic from multiple sources to make it unavailable.
SQL Injection	Inserting malicious SQL code into input fields to manipulate or extract database information.
Brute Force	Systematically trying all possible passwords or keys until the correct one is found.
Ransomware	Encrypts the victim's files and demands payment (ransom) for the decryption key.

1.3 Attack Vectors

- **Social Engineering:** Manipulating people psychologically to divulge confidential information or perform actions.
- **Wireless Attacks:** Exploiting Wi-Fi vulnerabilities such as WPA/WEP cracking, Evil Twin AP, and deauthentication floods.
- **Insider Threats:** Malicious or negligent actions by employees, contractors, or partners with authorized access.

A controlled, isolated lab environment is essential for practicing ethical hacking techniques safely without affecting real systems or networks.

2.1 VirtualBox Installation

- Download VirtualBox from https://www.virtualbox.org/wiki/Downloads
- Run the installer and follow the on-screen setup wizard.
- Install VirtualBox Extension Pack for USB and networking support.

2.2 Kali Linux Setup (Attacker Machine)

- Download the pre-built Kali Linux VirtualBox OVA from https://www.kali.org/get-kali/#kali-virtual-machines
- In VirtualBox: File → Import Appliance → select the .ova file.
- Default login credentials: Username: kali | Password: kali
- Run: sudo apt update && sudo apt upgrade -y to update all packages.

2.3 Metasploitable2 Setup (Target Machine)

- Download Metasploitable2 from https://sourceforge.net/projects/metasploitable/
- Extract the .zip and import the .vmdk into VirtualBox as a new VM.
- Default credentials: Username: msfadmin | Password: msfadmin
- This VM is intentionally vulnerable — never expose it to the internet.

■ *Security Note: Always keep both VMs on Host-Only network. Never bridge them to your real network.*

2.4 Network Configuration (Host-Only Adapter)

- Step 1: VirtualBox → File → Host Network Manager → click Create.
- Step 2: Note the network address (e.g., 192.168.56.0/24).
- Step 3: For each VM: Settings → Network → Adapter 1 → Attached to: Host-Only Adapter.
- Step 4: Boot Kali and run: ip a — note the assigned IP address.
- Step 5: Boot Metasploitable2 and run: ifconfig — note its IP.
- Step 6: From Kali, ping Metasploitable2: ping 192.168.56.101

2.5 Lab Architecture Diagram

Component	OS	Role	IP (Example)
Host Machine	Windows/macOS/Linux	Runs VirtualBox Hypervisor	192.168.56.1
Kali Linux VM	Kali Linux 2024	Attacker Machine	192.168.56.102
Metasploitable2 VM	Ubuntu 8.04	Vulnerable Target	192.168.56.101

Linux is the primary operating system used in cybersecurity. Kali Linux is Debian-based and mastering these commands is essential for all penetration testing tasks.

3.1 File System Navigation

```
pwd # Print current working directory ls # List files in current
directory ls -la # List all files with permissions & hidden
files cd /home/kali # Change to specified directory cd .. # Go
up one directory level mkdir myfolder # Create a new directory
rm -rf myfolder # Remove directory and contents (use carefully!)
cp file.txt /tmp/ # Copy file to /tmp/ mv file.txt newname.txt #
Rename or move a file
```

3.2 File & Directory Permissions

Linux permissions are represented as three sets of rwx (read/write/execute) for owner, group, and others. The chmod command modifies these permissions.

```
ls -l file.txt # View permissions: -rwxr-xr-x chmod 755
script.sh # rwx for owner, rx for group & others chmod 600
secret.txt # rw for owner only (private files) chmod +x
script.sh # Add execute permission chown kali:kali file #
Change owner and group sudo chown root file # Change owner
to root
```

3.3 Package Management

```
sudo apt update # Refresh package list from repositories
sudo apt upgrade -y # Upgrade all installed packages
sudo apt install nmap # Install a specific package sudo
apt remove nmap # Remove a package dpkg -l # List all
installed packages dpkg -l | grep nmap # Search for a
specific package
```

3.4 Networking Commands

```
ifconfig # Show all network interfaces and IPs ip a #
Modern alternative to ifconfig ping 192.168.56.101 #
Test connectivity to a host ping -c 4 google.com # Send
only 4 ping packets netstat -tulnp # Show open ports and
listening services ss -tulnp # Modern alternative to
netstat traceroute google.com # Trace the network path
to a host route -n # Show the routing table cat
/etc/resolv.conf # View DNS server configuration
```

4.1 The OSI Model

The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes network communication into 7 layers.

Layer	Name	Function	Protocols / Examples

7	**Application**	User interface and network services	HTTP, HTTPS, FTP, DNS, SMTP
6	**Presentation**	Data encryption, compression, translation	SSL/TLS, JPEG, ASCII
5	**Session**	Establishes and manages sessions	NetBIOS, RPC, PPTP
4	**Transport**	End-to-end data delivery and error checking	TCP, UDP
3	**Network**	Routing and IP addressing	IP, ICMP, OSPF, BGP
2	**Data Link**	MAC addressing and frame delivery	Ethernet, Wi-Fi, ARP
1	**Physical**	Physical transmission of bits	Cables, Fiber, Wi-Fi signals

4.2 Key Protocols

Protocol	Port	Purpose
HTTP	80	Unencrypted web browsing
HTTPS	443	Encrypted web browsing (TLS)
FTP	21	File transfer (unencrypted)
SSH	22	Secure remote shell access
DNS	53	Domain name to IP resolution
SMTP	25	Email sending
Telnet	23	Remote access (unencrypted — avoid)

4.3 IP Addressing & Subnetting

- **IPv4 Address:** 32-bit address written as four octets (e.g., 192.168.1.10).
- **Subnet Mask:** Defines the network portion. /24 = 255.255.255.0 = 256 hosts.
- **NAT (Network Address Translation):** Allows multiple private IPs to share one public IP.
- **Common Private Ranges:** 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16

5.1 Encryption Types

Type	Description	Example Algorithms	Use Case
Symmetric	Same key used for encryption and decryption.	`AES, DES, 3DES`	File encryption, disk encryption
Asymmetric	Public key encrypts; private key decrypts.	`RSA, ECC, Diffie-Hellman`	SSL/TLS, email encryption (PGP)
Hashing	One-way function — cannot be reversed.	`MD5, SHA1, SHA256, bcrypt`	Password storage, file integrity

5.2 Hands-On: OpenSSL Commands (Run in Kali Terminal)

```
# Generate SHA256 hash of a string
echo "Hello Internship" | sha256sum

# Generate MD5 hash echo "Hello
Internship" | md5sum

# Encrypt a file using AES-256-CBC openssl enc -aes-256-cbc -salt -in
plaintext.txt -out encrypted.enc -k mypassword

# Decrypt the encrypted file openssl enc -aes-256-cbc -d -in encrypted.enc -
out decrypted.txt -k mypassword

# Generate RSA private key (2048-bit)
openssl genrsa -out private.pem 2048

# Extract the public key from the private key
openssl rsa -in private.pem -pubout -out public.pem

# View certificate information openssl x509
-in certificate.crt -text -noout
```

5.3 Digital Certificates & SSL/TLS

- **SSL/TLS:** Protocol that encrypts data in transit between client and server (used in HTTPS).
- **Certificate Authority (CA):** Trusted entity that issues digital certificates (e.g., Let's Encrypt, DigiCert).
- **Digital Certificate:** Contains the public key and identity information, signed by a CA.
- **Handshake Process:** Client and server negotiate cipher suite, exchange certificates, and derive session keys.

6.1 Wireshark – Packet Capture

Wireshark is a network protocol analyzer used to capture and inspect packets in real time. It is invaluable for analyzing network traffic and detecting anomalies.

- Open Wireshark in Kali: Applications → Sniffing & Spoofing → Wireshark.
- Select the network interface (e.g., eth0) and click the blue shark fin to start capture.
- Apply display filters to focus on specific traffic.

```
http # Filter only HTTP traffic ftp # Filter FTP
traffic dns # Filter DNS queries tcp.port == 80 #
Filter traffic on port 80 ip.addr ==
192.168.56.101 # Filter by specific IP
```

6.2 Nmap – Network Scanner

Nmap (Network Mapper) is the industry standard tool for network discovery and security auditing. It can detect open ports, running services, and OS versions.

```
# Scan a single host
nmap 192.168.56.101

# Scan entire subnet
nmap 192.168.56.0/24

# SYN Stealth Scan (most common)
nmap -sS 192.168.56.101

# Service & Version Detection
nmap -sV 192.168.56.101

# OS Detection nmap -
O 192.168.56.101

# Aggressive scan (OS + Version + Scripts + Traceroute)
nmap -A 192.168.56.101

# Save output to file nmap -sV
192.168.56.101 -oN scan_report.txt
```

6.3 Burp Suite – Web Application Proxy

- Open Burp Suite Community in Kali: Applications → Web Application Analysis → Burp Suite.
- Go to Proxy → Options → confirm listener is on 127.0.0.1:8080.
- Configure Firefox: Preferences → Network Settings → Manual Proxy → 127.0.0.1 port 8080.
- Enable intercept in Burp Proxy tab to capture and modify HTTP requests.
- Use the Repeater tab to manually replay and modify requests.

6.4 Netcat – Network Swiss Army Knife

```
# Listen on a port (server mode)
nc -lvp 4444

# Connect to a host and port (client mode)
nc 192.168.56.101 4444

# Banner grabbing (identify service)
nc 192.168.56.101 80

# Transfer a file
# Receiver: nc -lvp 4444 > received_file.txt
# Sender: nc 192.168.56.102 4444 < file_to_send.txt
```

7. DELIVERABLES SUMMARY

The following deliverables are required for Task 1 submission. Each item must be completed and uploaded as per the ApexPlanet internship submission process.

#	Deliverable	Details	Format
1	**Lab Setup Report**	Screenshots of Kali Linux, Metasploitable2, and Wireshark capture	*PDF / DOCX*
2	**GitHub Repository**	Notes, Linux cheat-sheet, networking notes, OpenSSL output screenshots	*GitHub Repo (Public)*
3	**Video Walkthrough**	5-minute screen recording showing lab setup, ping test, Nmap scan, Wireshark capture	*LinkedIn Featured Video*

7.1 Submission Checklist

- VirtualBox installed and running on your machine
- Kali Linux VM imported and boots successfully
- Metasploitable2 VM imported and boots successfully
- Both VMs set to Host-Only network adapter
- Successful ping from Kali to Metasploitable2
- Nmap scan run against Metasploitable2 (screenshot saved)
- Wireshark opened and packet capture demonstrated
- OpenSSL encryption/decryption commands executed with screenshots
- GitHub repo created with notes and cheat sheet
- 5-minute screen recording uploaded to LinkedIn Featured section
- Task submitted on ApexPlanet portal with LinkedIn + GitHub links

This report was prepared as part of the ApexPlanet Software Pvt. Ltd. Cybersecurity & Ethical Hacking Internship Program. All practical exercises were conducted in a safe, isolated lab environment.